

Da Wei

Embodied AI • Vision–Language–Action Learning • World Models

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Education

University of Washington, Seattle

B.S. in Informatics GPA: 3.73/4.0

Expected 2027

Seattle, WA

The Chinese University of Hong Kong

Exchange Student

Fall 2026

Hong Kong SAR

Research Experience

Multimodal Intelligence Lab, University of Washington

Undergraduate Research Assistant • PI: Zhi-Qi Cheng

Apr 2026–Present

Seattle, WA

Human-Aware Vision–Language Navigation

Ongoing research

- Built a batch 3D motion-retargeting agent with Kimodo and Virtual Community for new VLN scenes; automated data preparation and audited and repaired **300 human-motion samples**.
- Trained and evaluated **five VLN models with DAGger-based imitation learning** in HA-VLN 3.0's DE environment: Speaker-Follower, Rec-PREVALENT, Rec-OSCAR, AirBERT, and HAMT.
- Improved success rate across all five models by **6.15 percentage points on average**, with lower navigation error and collision metrics after training.

Multimodal Speculative Decoding

Ongoing research

- Analyzed limitations of the ViSpec baseline and formulated an objective linking visual-compression quality with draft–target model behavior.
- Proposed autoencoder supervision to measure and reduce visual-adaptor information loss.

Independent Research

MATH-LWM-AGENT: Efficient Mathematical Tool Agent

Sole author • Under review at AAAI 2027

- **Conceived and developed a training-free mathematical agent** that predicts tool outcomes and uses prediction residuals to guide reasoning and tool use.
- Implemented runtime and residual control with Python, SymPy, and verification tools.
- Led experiment design, ablations, inference-cost accounting, evidence audits, and paper preparation.
- With thinking disabled, improved HMMT by **52.13 pp** and IMO-AnswerBench by **35.85 pp**.

Research Experience continued

NUS LinS Lab & RoboScience Lab

Research Assistant • PI: Lin Shao

Aug 2026–Present

Shenzhen, China

WAQ: Unified World–Action–Critic Model

Sixth author • Planned submission

- **Owned the end-to-end real-robot pipeline** across four embodiments and five manipulation settings, covering data collection, curation, deployment, final evaluation, and demonstrations.
- Ran video–action supervised fine-tuning, then refined offline-RL critics with the backbone frozen, for xArm-gripper, bimanual xArm-gripper, and xHand with and without FingerEye.
- Evaluated WAQ against LingBot-VA; on xArm–xHand eggplant-to-cup, WAQ achieved **70% vs. 20% full-task success** and a **77.5% vs. 30% average grasp/placement score**.

CRL-VLA: Continual Vision–Language–Action Learning

Fourth author • Submitted to ICLR 2027

- **Owned the offline and online RL real-robot pipeline** on xArm-gripper: data collection, baseline and CRL-VLA policy training, deployment, and checkpoint-level evaluation.
- Contributed to core-method and real-robot experiment design for a continual-learning framework combining a frozen retention critic with a trainable adaptation critic.
- In five-task offline xArm evaluation, CRL-VLA achieved **58% versus 34% mean final task success** and **82% versus 60% mean grasp success** against the baseline.
- In three-task online xArm evaluation, CRL-VLA achieved a mean combined grasp and placement score of **0.35 versus 0.13**. Real-robot comparisons were descriptive single-run evaluations.

Open Source and Leadership

Wordplay

Open-Source Project Maintainer

Dec 2025–Jun 2026

Seattle, WA

- Contributed to Amy J. Ko’s educational programming language and IDE; reviewed accessibility and multilingual typography, and translated usability findings into implementation tasks.

“Xiao Wang Xuezhong” Education Media Channel

Founder and Content Creator

Jan 2026–Present

China / Online

- Built a WeChat Channels audience of approximately **5,000 followers**, sharing undergraduate application guidance and international-student experiences.

Research Methods and Skills

Robot learning Vision–language–action models, offline and online reinforcement learning, supervised fine-tuning, imitation learning, continual learning, and cross-embodiment evaluation.

Experimental systems Real-robot data collection and curation, policy deployment, checkpoint evaluation, ablation studies, and inference-cost analysis.

Programming and tools Python, Java, R, LaTeX, and SymPy.

Additional research LLM/VLM inference, speculative decoding, tool use, prediction-residual reasoning.